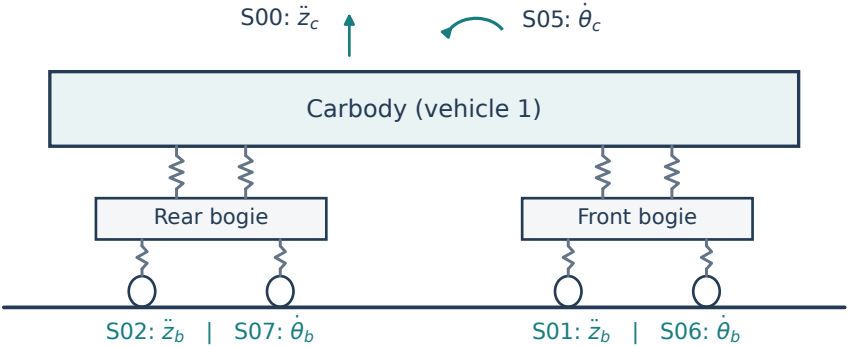
(a) Train and leading-vehicle response channels

Five planar vehicles; leading vehicle = 1

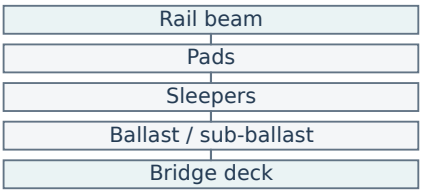


$\ddot{z}$: vertical acceleration | $\dot{\theta}$: pitch rate

S03/S04: constrained-wheelset acceleration proxies; diagnostic only

Four constrained wheelsets per vehicle; no independent wheel-motion DOFs

(b) Retained track layers and coupling



Layer inventory only

Inherited on-bridge ballast mass is condensed onto deck DOFs.

Adjacent-ballast shear coupling is omitted in this model.

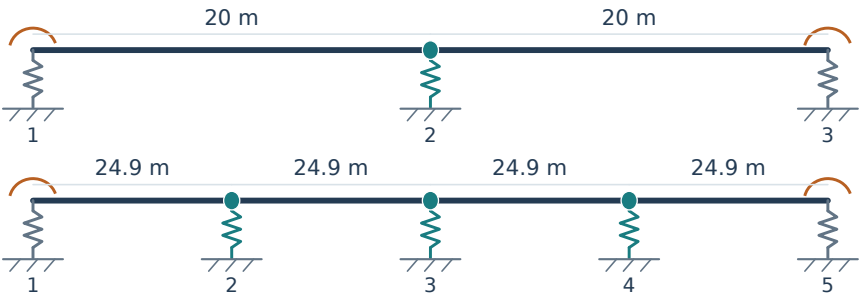
(c) Prospective regression bridge geometries

F40

40 m

L99

99.6 m



Teal: scour targets | Orange: abutment rotational springs in M blocks

F25 is separate: 2 × 19.95 m = 39.9 m. All panels are schematic, not to scale.